HW2

2.3 (a) Prove that the *Affine Cipher* achieves perfect secrecy if every key is used with equal probability $1/312$.

(b) More generally, suppose we are given a probability distribution on the set

$$\{a \in \mathbb{Z}_{26} : \gcd(a, 26) = 1\}.$$

Suppose that every key (a, b) for the *Affine Cipher* is used with probability $\mathbf{Pr}[a]/26$. Prove that the *Affine Cipher* achieves perfect secrecy when this probability distribution is defined on the keyspace.

(a) 任取一个固定明文 $m \in \mathbb{Z}_{26}$，一个固定密文 $c \in \mathbb{Z}_{26}$

有加密方程： $c \equiv am + b \pmod{26}$ ①

对于 $\mathbb{Z}_{26}^*$ 中任意一个 a，等价于 $b \equiv c - am \pmod{26}$ ②

m、c、a 均确定，所以有且仅有一个 $b \in \mathbb{Z}_{26}$ 满足 ②

又∵ 仿射密码中 $\forall K = (a, b) \in \mathcal{K}$ 等概率 ∴ $Pr(K = (a,b)) = \frac{1}{312}$

因此在 312 个 K 中，能够满足 ① 的 K 恰有 $|\mathbb{Z}_{26}^*| = 12$ 个

$Pr(C = c \mid M = m) = \sum Pr(K = K) = 12 \times \frac{1}{312} = \frac{1}{26}$

$Pr(C = c) = \sum Pr(C = c \mid M = m') \cdot Pr(M = m') = \sum \frac{1}{26} \cdot Pr(M = m')$

∵ 所有明文出现的概率和为 1 ∴ $\sum Pr(M = m') = 1$

∴ $Pr(C = c) = \frac{1}{26}$ 即 $\forall m \in \mathcal{M}, c \in \mathcal{C}$，有 $Pr(C = c \mid M = m) = Pr(C = c)$

满足完善保密性的定义，证毕.

(b) 任取一固定明文 $m \in \mathbb{Z}_{26}$ 与一固定密文 $c \in \mathbb{Z}_{26}$

与(a)同理 对 $\forall a \in \mathbb{Z}_{26}^*$，存在唯一的 $b \equiv c - am \pmod{26}$ ①

令满足 ① 的 $K = (a, b)$ 为集合 $K_{m,c}$

$Pr(C = c \mid M = m) = \sum_{(a,b) \in K_{m,c}} Pr(K = (a,b))$

有 $Pr(K = (a,b)) = \frac{Pr(a)}{26}$

$Pr(C = c \mid M = m) = \sum_{a \in \mathbb{Z}_{26}^*} \frac{Pr(a)}{26} = \frac{1}{26} \sum_{a \in \mathbb{Z}_{26}^*} Pr(a)$ ∵ $\sum_{a \in \mathbb{Z}_{26}^*} Pr(a) = 1$ ∴ $Pr(C = c \mid M = m) = \frac{1}{26}$

$Pr(C = c) = \sum_{m' \in \mathcal{M}} Pr(C = c \mid M = m') \cdot Pr(M = m') = \sum_{m' \in \mathcal{M}} \frac{1}{26} \cdot Pr(M = m') = \frac{1}{26}$

对 $\forall m \in \mathcal{M}, c \in \mathcal{C}$，仍有 $Pr(C = c \mid M = m) = Pr(C = c)$

证毕.